

- 6 The phenotype of an organism is the result of interactions between the genotype and the environment.

Phenotypes can show continuous or discontinuous variation in a population.

- (a) Sketch a graph to show the distribution for a phenotype showing continuous variation in a population.

(2)

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- (b) The presence or absence of teeth in chickens is one example of a discontinuous phenotype.

Chicken genomes contain the genes for tooth production, but chickens do not normally develop teeth.

These genes are switched off as the embryo develops.

Explain how a mutation could cause the development of teeth in a chicken embryo.

(4)

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(c) The photograph shows two Himalayan goats.



(Source: © Rainer Pauschert / Alamy Stock Photo)

The fur colour of a goat depends on its genotype.

Gene B controls fur colour in these goats.

The allele for brown fur is dominant to the allele for white fur in goats.

The allele **B** codes for brown fur and the allele **b** codes for white fur.

In one population, there were 470 brown goats and 140 white goats.

Calculate the frequency of the allele **B** in this population, using the equation

$$p^2 + 2pq + q^2 = 1$$

Give your answer to **two** decimal places.

(3)

Answer

(Total for Question 6 = 9 marks)

